MACROANALYSIS

Digital Methods & Literary History

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2 EVIDENCE

Scientists scoff at each other's theories but agree in basing them on the assumption that evidence, properly observed and measured, is true.

—Felipe Fernández-Armesto

While still graduate students in the early 1990s, my wife and I invited some friends to share Thanksgiving dinner. One of the friends was, like my wife and me, a graduate student in English. The other, however, was an outsider, a graduate student from geology. The conversation that night ranged over a wine-fueled spectrum of topics, but as three of the four of us were English majors, things eventually came around to literature. There was controversy when we came to discuss the "critical enterprise" and what it means to engage in literary research. The very term *research* was discussed and debated, with the lone scientist in the group suggesting, asserting, that the "methodology" employed by literary scholars was a rather subjective and highly anecdotal one, one that produced little in terms of "verifiable results" if much in the way of unsupportable speculation.

I recall rising to this challenge, asserting that the literary methodology was in essence no different from the scientific one: I argued that scholars of literature (at least scholars of the idealistic kind that I then saw myself becoming), like their counterparts in the sciences, should and do seek to uncover evidence and discover meaning, perhaps even truth. I dug deeper, arguing that literary scholars employ the same methods of investigation as scientists: we form a hypothesis about a literary work and then engage in a process of gathering evidence to test that hypothesis.

After so many years it is only a slightly embarrassing story. Although I am no longer convinced that the methods employed in literary studies are exactly the same as those employed in the sciences, I remain convinced that there are a good many methods worth sharing and that the similarities of methods exist in concrete ways, not simply as analogous practices.

The goal of science, we hope, is to develop the best possible explanation for some phenomenon. This is done via a careful and exhaustive gathering of evi-

dence. We understand that the conclusions drawn are only as good as the evidence gathered, and we hope that the gathering of evidence is done both ethically and completely. If and when new evidence is discovered, prior conclusions may need to be revised or abandoned—such was the case with the Ptolemaic model of a geocentric universe. Science is flexible in this matter of new evidence and is open to the possibility that new methods of investigation will unearth new, and sometimes contradictory, evidence.

Literary studies should strive for a similar goal, even if we persist in a belief that literary interpretation is a matter of opinion. Frankly, some opinions are better than others: better informed, better derived, or just simply better for being more reasonable, more believable. Science has sought to derive conclusions based on evidence, and in the ideal, science is open to new methodologies. Moreover, to the extent possible, science attempts to be exhaustive in the gathering of the evidence and must therefore welcome new modes of exploration, discovery, and analysis. The same might be said of literary scholars, excepting, of course, that the methods employed for the evidence gathering, for the discovery, are rather different. Literary criticism relies heavily on associations as evidence. Even though the notions of evidence are different, it is reasonable to insist that some associations are better than others.

The study of literature relies upon careful observation, the sustained, concentrated reading of text. This, our primary methodology, is "close reading." Science has a methodological advantage in the use of experimentation. Experimentation offers a method through which competing observations and conclusions may be tested and ruled out. With a few exceptions, there is no obvious corollary to scientific experimentation in literary studies. The conclusions we reach as literary scholars are rarely "testable" in the way that scientific conclusions are testable. And the conclusions we reach as literary scholars are rarely "repeatable" in the way that scientific experiments are repeatable. We are highly invested in interpretations, and it is very difficult to "rule out" an interpretation. That said, as a way of enriching a reader's experience of a given text, close reading is obviously fruitful; a scholar's interpretation of a text may help another reader to "see" or observe in the text elements that might have otherwise remained latent. Even a layman's interpretations may lead another reader to a more profound, more pleasurable understanding of a text. It would be wasteful and futile to debate the value of interpretation, but interpretation is fueled by observation, and as a method of evidence gathering, observation—both in the sciences and in the humanities—is flawed. Despite all their efforts to repress them, researchers will have irrepressible biases. Even scientists will "interpret" their evidence through a lens of subjectivity. Observation is flawed in the same way that generalization from the specific is flawed: the generalization may be good, it may even explain a total population, but the selection of the sample is always something less than

perfect, and so the observed results are likewise imperfect. In the sciences, a great deal of time and energy goes into the proper construction of "representative samples," but even with good sampling techniques and careful statistical calculations, there remain problems: outliers, exceptions, and so on. Perfection in sampling is just not possible.

Today, however, the ubiquity of data, so-called big data, is changing the sampling game. Indeed, big data are fundamentally altering the way that much science and social science get done. The existence of huge data sets means that many areas of research are no longer dependent upon controlled, artificial experiments or upon observations derived from data sampling. Instead of conducting controlled experiments on samples and then extrapolating from the specific to the general or from the close to the distant, these massive data sets are allowing for investigations at a scale that reaches or approaches a point of being comprehensive. The once inaccessible "population" has become accessible and is fast replacing the random and representative sample.

In literary studies, we have the equivalent of this big data in the form of big libraries. These massive digital-text collections—from vendors such as Chadwyck-Healey, from grassroots organizations such as Project Gutenberg, from nonprofit groups such as the Internet Archive and HathiTrust, and from the elephants in Mountain View, California, and Seattle, Washington*—are changing how literary studies get done. Science has welcomed big data and scaled its methods accordingly. With a huge amount of digital-textual data, we must do the same. Close reading is not only impractical as a means of evidence gathering in the digital library, but big data render it totally inappropriate as a method of studying literary history. This is not to imply that scholars have been wholly unsuccessful in employing close reading to the study of literary history. A careful reader, such as Ian Watt, argues that elements leading to the rise of the novel could be detected and teased out of the writings of Defoe, Richardson, and Fielding. Watt's study is magnificent; his many observations are reasonable, and there is soundness about them.† He appears correct on a number of points, but he has observed only a small space. What are we to do with the other three to five thousand works of

* That is, Google.com and Amazon.com.

† A similar statement could be made of Erich Auerbach's *Mimesis.* It is a magnificent bit of close reading. At the same time, Auerbach was acutely aware of the limitations of his methodology. In the epilogue to *Mimesis,* he notes the difficulties of dealing with "texts ranging over three thousand years" and how the limitations of his library in Istanbul made it "probable that [he] overlooked things which [he] ought to have considered." Interestingly, however, he says at the same time that "it is quite possible that the book owes its existence to just this lack of a rich . . . library." If it had been possible to access the greater archive, he "might never have reached the point of writing" (1953).

fiction published in the eighteenth century? What of the works that Watt did not observe and account for with his methodology, and how are we to now account for the works not penned by Defoe, by Richardson, or by Fielding? Might other novelists tell a different story? Can we, in good conscience, even believe that Defoe, Richardson, and Fielding are representative writers? Watt's sampling was not random; it was quite the opposite. But perhaps we only need to believe that these three (male) authors are representative of the trend toward "realism" that flourished in the nineteenth century. Accepting this premise makes Watt's magnificent synthesis into no more than a self-fulfilling project, a project in which the books are stacked in advance. No matter what we think of the sample, we must question whether in fact realism really did flourish. Even before that, we really ought to define what it means "to flourish" in the first place. Flourishing certainly seems to be the sort of thing that could, and ought, to be measured. Watt had no such yardstick against which to make a measurement. He had only a few hundred texts that he had read. Today, things are different. The larger literary record can no longer be ignored: it is here, and much of it is now accessible.

At the time of my Thanksgiving dinner back in the 1990s, gathering literary evidence meant reading books, noting "things" (a phallic symbol here, a biblical reference there, a stylistic flourish, an allusion, and so on) and then interpreting: making sense and arguments out of those observations.* Today, in the age of digital libraries and large-scale book-digitization projects, the nature of the "evidence" available to us has changed, radically. Which is not to say that we should no longer read books looking for, or noting, random "things," but rather to emphasize that massive digital corpora offer us unprecedented access to the literary record and invite, even demand, a new type of evidence gathering and meaning making. The literary scholar of the twenty-first century can no longer be content with anecdotal evidence, with random "things" gathered from a few, even "representative," texts.† We must strive to understand these things we find interesting in the context of everything else, including a mass of possibly "uninteresting" texts.

* Yes, a simplification, but close enough to serve as a heady foil in this introductory polemic. Along similar lines, Susan Hockey writes of the "somewhat serendipitous noting of interesting features" (2000, 66).

† When writing of "anecdotal" here, I am not thinking of the use made of anecdote in the new historical tradition that we find expressed in, for example, Greenblatt's "cultural poetics." Rather, I use the word in the sense of "anecdotal evidence": that is, evidence that is atypical, informally gathered, speculative, or purely interpretive, which is to say not empirical. On this point, the type of literary data I am exploring allows me to adopt a fundamentally empirical position. Having said that, there is a place for anecdotal evidence in literary study, and I do not intend here a critique of anecdotalism per se, but rather to simply make a distinction and separation between two types of evidence.

"Strictly speaking," wrote Russian formalist Juri Tynjanov in 1927, "one cannot study literary phenomena outside of their interrelationships" (1978, 71). Unfortunately for Tynjanov, the multitude of interrelationships far exceeded his ability to study them, especially with close and careful reading as his primary tools. Like it or not, today's literary-historical scholar can no longer risk being *just* a close reader: the sheer quantity of available data makes the traditional practice of close reading untenable as an exhaustive or definitive method of evidence gathering. Something important will inevitably be missed. The same argument, however, may be leveled against the macroscale; from thirty thousand feet, something important will inevitably be missed. The two scales of analysis, therefore, should and need to coexist. For this to happen, the literary researcher must embrace new, and largely computational, ways of gathering evidence. Just as we would not expect an economist to generate sound theories about the economy by studying a few consumers or a few businesses, literary scholars cannot be content to read literary history from a canon of a few authors or even several hundred texts. Today's student of literature must be adept at reading and gathering evidence from individual texts and equally adept at accessing and mining digital-text repositories. And *mining* here really is the key word in context. Literary scholars must learn to go beyond search. In search we go after a single nugget, carefully panning in the river of prose. At the risk of giving offense to the environmentalists, what is needed now is the literary equivalent of open-pit mining or hydraulicking. We are proficient at electronic search and comfortable searching digital collections for some piece of evidence to support an argument, but the sheer amount of data now available makes search ineffectual as a means of evidence gathering. Close reading, digital searching, will continue to reveal nuggets, while the deeper veins lie buried beneath the mass of gravel layered above. What are required are methods for aggregating and making sense out of both the nuggets and the tailings. Take the case of a scholar conducting research for a hypothetical paper about Melville's metaphysics. A query for *whale* in the Google Books library produces 33,338 hits—way too broad. Narrowing the search by entering *whale* and *god* results in a more manageable 3,715 hits, including such promising titles as *American Literature in Context* and *Melville's Quarrel with God.* Even if the scholar could further narrow the list to 1,000 books, this is still far too many to read in any practical way. Unless one knows what to look for—say, a quotation only partially remembered—searching for research purposes, as a means of evidence gathering, is not terribly practical.* More interesting, more exciting, than panning for nuggets in digital archives

* In revising this section before publication, I went back to Google Books and discovered that the number of hits for this particular search had grown significantly since I first tested. No doubt readers will find even higher numbers today.

is the ability to go beyond the pan and exploit the trommel of computation to process, condense, deform, and analyze the deeper strata from which these nuggets were born, to unearth, for the first time, what these corpora really contain. In practical terms, this means that we must evolve to embrace new approaches and new methodologies designed for accessing and leveraging the electronic texts that make up the twenty-first-century digital library.

This is a book about evidence gathering. It is a book about how new methods of analysis allow us to extract new forms of evidence from the digital library. Nevertheless, this is also a book about literature. What matter the methods, so long as the results of employing them lead us to a deeper knowledge of our subject? A methodology is important and useful if it opens new doorways of discovery, if it teaches us something new about literary history, about individual creativity, and about the seeming inevitability of influence.